

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

Department of Computer Science and Engineering

**CSE-2340: Software Development 1**

# Design Handoff & CSS Positioning

*Week 4 · Class 1 (teaching) + Class 2 (practice) · Figma/Pixso, position, z-index*

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# Recall & Today's Outcome

## ? Recall from Week 3

- Which property spreads items along the main axis?
- What are the three questions the cascade asks?
- Why is margin: 0 auto not centring my box?
- How did MP1 go?

## ✓ By the end of today you can

- Read a design file and take real measurements from it
- Export assets at the right size and format
- Use relative, absolute, fixed and sticky correctly
- Explain what a positioning context is
- Use z-index without guessing numbers
- Build a design given to you, not invented by you

# Why Design Handoff Matters

## How students usually work

- Invent the layout while coding it
- Pick colours and spacing by feel
- Every page ends up slightly different
- "It looks about right"

## How a job works

- A designer gives you a file
- You read exact values from it
- You reproduce it, you do not redesign it
- "It matches the design"

## Your first front-end job will start with a Figma link, not a blank file

- Reading a design is a skill on its own: finding spacing, type sizes, colours and asset sizes without asking the designer.
- Mini Project 2 in Week 5 asks you to build your OWN design, so today you learn both sides: how to read one and how to record one.

# Figma or Pixso: What You Need

## Frame

One screen or one component. The artboard you design inside.

## Layers panel

The tree on the left. It maps almost one to one onto your HTML structure.

## Inspect / Dev mode

The panel on the right. Select anything and read its real values.

## Distance measuring

Select one element, hold Alt (Option), hover another. The gap appears.



## Pixso works the same way and is often faster in Bangladesh

Either tool is fine for this course. The panels have different names but the same four things: layers, measurements, styles and export.

# What to Read From a Design

## Spacing

Padding inside, margin between. Measure, do not guess.

24px, 16px

## Type

Family, size, weight, line height.

Arial 16 / 1.6

## Colour

Copy the hex value. Do not eyedrop from a screenshot.

#003366

## Radius and border

Corner radius and border width.

8px, 1px

## Size

Fixed width, or does it stretch?

max-width 1100px

## Breakpoints

Is there a mobile frame as well as a desktop one?

360 / 768 / 1280

# Exporting Assets

## SVG

Logos, icons, simple shapes

*Scales perfectly, tiny file*

## PNG

Images needing transparency

*Export at 2x for sharp screens*

## JPG

Photographs

*Smaller than PNG, no transparency*

## WebP

Photographs, modern sites

*Smallest. Supported everywhere now*

## ! Why 2x

- A phone screen packs two or three real pixels into one CSS pixel
- A 1x image looks soft on it
- Export at 2x, then set the CSS width to the 1x value

## ✓ Housekeeping

- Lowercase file names, hyphens not spaces
- Keep them in an images/ folder
- Compress photos before committing them

# 01

## CSS Positioning

Flexbox and Grid arrange the flow. Positioning takes an element out of it.

# The Five Values of position

## **static**

The default. The element stays in normal flow. top/left do nothing.

## **relative**

Stays in flow and keeps its space, but can be nudged. Also creates a positioning context.

## **absolute**

Removed from flow. Positioned against the nearest positioned ancestor.

## **fixed**

Removed from flow. Positioned against the viewport. Does not scroll away.

## **sticky**

Normal until it hits a scroll threshold, then it sticks.

# The Rule: relative Parent, absolute Child

```
/* The parent creates the context */  
.card {  
  position: relative;  
}  
  
/* The child is placed inside THAT box */  
.card__badge {  
  position: absolute;  
  top: 12px;  
  right: 12px;  
}
```

**position: relative**

NEW

Card content sits here

*top: 12px right: 12px → measured from the card, not the page*

## The bug you will hit

- Forget position: relative on the parent and the badge flies to the corner of the PAGE, not the card.
- An absolutely positioned element looks up through its ancestors for the first one that is positioned. If it finds none, it uses the whole document.
- So the question is never "where did my badge go" but "which ancestor is it measuring from?"

# sticky and fixed in Practice

```
/* Sticks once it reaches the top of the screen */  
.site-header {  
  position: sticky;  
  top: 0;  
  z-index: 10;  
}  
  
/* Always visible, ignores scrolling entirely */  
.back-to-top {  
  position: fixed;  
  bottom: 20px;  
  right: 20px;
```

## sticky vs fixed

- sticky keeps its space in the flow; fixed does not
- sticky stops at the end of its parent; fixed never stops
- A sticky header pushes content down; a fixed one overlaps it

## ! sticky needs two things

- A threshold: top, bottom, left or right
- A parent that is tall enough to scroll
- Without top: 0 it does nothing at all

## Use sparingly

- A fixed header on a phone eats a third of a small screen
- Two sticky bars are one too many
- If it covers content, add padding to the body

# z-index Without Guessing

```
/* z-index only works on a POSITIONED element */
.badge {
  position: absolute; /* required */
  z-index: 2;
}

/* Agree a scale and stick to it */
/* 10 sticky header
   20 dropdown
   30 modal overlay
   40 toast message */
```



## Two rules people forget

- z-index does nothing without position (other than in flex or grid)
- A parent with its own z-index traps its children: a child can never escape above its parent's level

## Why 9999 is a bad habit

- The next element needs 10000, then 99999
- Nobody can tell what is meant to sit above what
- Use a small documented scale instead



## Debug it

- If z-index does nothing, check for position first
- Then check whether an ancestor has its own z-index
- DevTools shows the computed value

# When NOT to Use Positioning

## Do not position for layout

- Two columns side by side → Flexbox or Grid
- Centring a box → Flexbox, or margin: 0 auto
- A row of cards → Flexbox
- Absolute positioning does not respond to screen size

## Do position for overlays

- A badge on a card corner
- A caption over an image
- A sticky header
- A back-to-top button
- A modal dialog

## The test question

- Does this element need to sit ON TOP of something else? Then position it.
- Does it need to sit NEXT TO something else? Then use Flexbox or Grid.
- Remember Week 2: absolute positioning was the painful way to place a button. Do not go back to it.

# Live Build: Sticky Header and Card Badge

```
/* 1. Sticky header */
.site-header {
  position: sticky;
  top: 0;
  z-index: 10;
  background-color: var(--navy);
  /* MUST be opaque, or the page
     text scrolls through it */
}

/* 2. The positioning context */
.card {
  position: relative;
  overflow: hidden;
}

/* 4. Caption over the image */
.card__caption {
  position: absolute;
  bottom: 0;
  left: 0;
  right: 0; /* left + right = full width */
  background-color: rgba(0,0,0,0.55);
  color: #FFFFFF;
  padding: 10px;
}
```

```
/* 3. Badge in the corner */
.card__badge {
  position: absolute;
  top: 12px;
  right: 12px;
  z-index: 2;
  background-color: var(--amber);
  padding: 4px 10px;
  border-radius: 999px;
}
```



## Build and test one at a time

- After each block: scroll the page, then resize to 360px
- Three changes at once means three interacting bugs

# Common Mistakes

**My badge flew to the page corner**

The parent has no position: relative. Absolute looks up for the nearest positioned ancestor.

**top / left do nothing**

The element is still position: static. Those properties are ignored.

**sticky is not sticking**

No top value, or an ancestor has overflow: hidden.

**z-index is ignored**

The element is not positioned, or an ancestor's z-index traps it.

**Content scrolls through my header**

The sticky header has a transparent background. Give it an opaque one.

**It breaks on a phone**

You positioned something that should have been laid out with Flexbox or Grid.

*Exit questions for C1: what does position: relative do that static does not? And why does z-index sometimes do nothing?*

# 02

## Class 2 — Practice

You are given a design. Build it exactly, then implement the positioning requirements.

# Class 2: What You Will Do

## 1 Read the design

- Take six measurements from the handoff file
- 8 min

## 2 Build the card

- Match the spacing, type and colours exactly
- 12 min

## 3 Add the positioned parts

- Badge, caption, sticky header
- 10 min

## 4 Fix the broken page

- Six positioning bugs planted in starter/
- 8 min

## ! The rule for today: reproduce, do not redesign

If you think the design is wrong, note it and build it anyway. Arguing with the design is a conversation for the designer, not a licence to change it.

# Exercise 1: Read the Design

Open the shared design file. Write these six values into measurements.md before writing any CSS.

## 1 Card padding

Space inside the card border

## 2 Gap between cards

Measure with Alt / Option hover

## 3 Heading size and weight

In px, plus the line height

## 4 Body text colour

Copy the hex, do not eyedrop

## 5 Corner radius

On the card and on the badge

## 6 Badge offset

Distance from the card's top and right edges

**Write the values down first. Coding while measuring is how you end up with 23px padding in one place and 25px in another.**

## Exercise 2: Build the Card

### Requirements

- Semantic markup: article, h3, p, a
- Use YOUR measured values, not invented ones
- Colours as custom properties in :root
- Three cards in a wrapping Flexbox row



### Do not

- Position anything yet — that is Exercise 3
- Change the colours because you prefer others
- Round 23px up to 24px without noting it

```
:root {  
  --card-pad: /* your measured value */;  
  --card-gap: /* your measured value */;  
  --radius:   /* your measured value */;  
}
```

# Exercise 3: The Positioned Parts

## 1 Badge on the card corner

position: relative on the card, absolute on the badge, at your measured offset

## 2 Caption over the image

Absolute, pinned to bottom/left/right, with a semi-transparent background

## 3 Sticky header

sticky with top: 0, an opaque background, and a sensible z-index

**Check after each one: scroll the page, resize to 360px, and Tab through it. Do not leave all three to the end.**

# Exercise 4: Fix the Broken Page

Open starter/. Six positioning bugs are planted in the CSS. The HTML is correct.

## 1 The badge sits in the page corner

Which ancestor is it measuring from?

## 3 The header does not stick

Two possible causes. Check both

## 5 The caption is behind the image

One property, and it needs a friend

## 2 top and left do nothing on the tag

Check the position value first

## 4 Page text scrolls through the header

Look at the header's background

## 6 z-index: 9999 still does not work

An ancestor is trapping it

**Same method as always: Inspect, read the computed position, find the rule, test live, then edit the file.**

# Assignment - Week 4, C1 and C2

## 1 Part 1 - In-class demonstration

- Time: last 5 minutes of C2
- Task: show your card with a correctly positioned badge, and your working sticky header
- Scroll the page and resize to 360px in front of the instructor
- Show your measurements.md next to the design file
- Marks: 2 (checked on the spot)

## 2 Part 2 - Home task

- Finish all four exercises including the six bug fixes
- Add a back-to-top button using position: fixed
- Write handoff-notes.md: your six measurements, plus anything in the design you could not reproduce and why
- Commit and push
- Deadline: before Week 4, Class 3 · Marks: 3

**Next class starts Tailwind CSS v4. Mini Project 2 (your own responsive design in Tailwind) is due Week 5, C4.**

# Marking Rubric & Exit Check

## RUBRIC - 5 MARKS TOTAL

|                                                         |     |
|---------------------------------------------------------|-----|
| Six measurements recorded and matched in the CSS        | 1.5 |
| Badge and caption correctly positioned                  | 1.5 |
| Sticky header works, with opaque background and z-index | 1.0 |
| All six planted bugs fixed                              | 1.0 |

*Next class (C3): Tailwind CSS v4 — the same Flexbox, Grid and spacing you already know, with shorter names.*

### Exit question 1 - concept

An absolutely positioned element measures its top and left from what, exactly?

### Exit question 2 - practical

Write the two declarations that make a header stick to the top of the screen.